

Aliah University
Even-Semester (Autumn) Examination - 2025
(4th Semester BCA)

Paper Name: Formal Language and Automata Theory
Paper Code: CSAUGPC20

Full Marks: 80
Time: 3 Hrs

Group-A

1. Answer all

5x2 = 10

- (i) Which data structure is used in case of a pushdown automata? [CO1]
- (ii) What are the limitations of finite automata? [CO1]
- (iii) What is the type of regular grammar as per chomsky hierarchy? [CO2] *19/23*
- (iv) What is a finite automata with output? [CO2]
- (v) What do you mean by configuration of a pushdown automata? [CO3]

Group-B

2. Answer any five

5x6 = 30

- (i) Specify the properties of DFA and NFA. [CO2]
- (ii) Design a finite automata which accepts all strings over {a,b} that end with substring "aba" [CO2]
- (iii) What is Kleene's theorem? Why is the pumping lemma used? [CO4]
- (iv) Design a pushdown automata which accepts a set of all palindromes over {0,1}. [CO2]
- (v) With an example show that non deterministic pushdown automata are more powerful than deterministic pushdown automata. [CO3]
- (vi) Find the regular expression: Set of all strings over {0,1} which have exactly one '0'. [CO2]
- (vii) Why are context free languages called "context free"? [CO2]

Group-C

(Answer any four)

3. Answer the followings:

8+2

- (a) Discuss about the Chomsky hierarchy of formal languages. [CO2]
- (b) Show that string concatenation is associative. [CO1]

4. Answer the followings:

6+4

- (a) Draw the block diagram of a finite automata and discuss different components. [CO2]
- (b) Define NFA and DFA. [CO2]

5. Write short notes (any two):

5x2

- (a) Acceptance criteria of a pushdown automata [CO3]
- (b) Regular grammar and context free grammar comparison [CO2]
- (c) Turing machine [CO2]

6. Answer the followings:

5+5

- (a) Design a pushdown automata which accepts all strings over $\{a,b\}$ where the number of 'a' is more than the number of 'b' in each string. [CO3] ~~$a^n b^n$~~ $a^{2n} b^n$
- (b) Design a DFA which accepts all strings over $\{a,b\}$ that starts with 'a' and ends with 'b'. Convert the same DFA into equivalent NFA. [CO3]

7. Explain (any two)

5x2

- (a) Writing in the input tape can make a powerful model of computation. [CO4]
- (b) The idea of modern computer starts with Turing machine [CO4]
- (c) DFA and NFA are equivalent in acceptance power [CO2]
- (d) Regular language, regular expression, finite automata are equivalent [CO2]

- 1) b
- 2) a
- 3) b
- 4) c
- 5) c
- 6) a
- 7) b
- 8) c
- 9) b
- 10) d